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Next item →

1. Which of the following assumptions did we make when deriving the prediction steps of the linear Kalman filter in this lesson? Select all that apply.

1 / 1 point

☒ We assumed that both the process noises and sensor noises experienced by the system are white noises.

☒ **Correct**
Yes. We need both the zero-mean and self-uncorrelated aspects of white noise for the derivation to hold.

☒ We assumed that the process and sensor noises experienced by the system are mutually uncorrelated.

☒ **Correct**
Yes. It is possible to rederive the Kalman filter if these noises are correlated with each other (e.g., if both noises arise in part from a common source), but we did not explore this generalization in this lesson.

☐ We assumed that the system whose state we are estimating is represented in a **time-invariant** state-space form.

☒ We assumed that the system whose state we are estimating is represented in a **linear** state-space form.

☒ **Correct**
Yes. The Kalman filter is derived for linear systems only.

☒ We assumed that both the process noises and sensor noises experienced by the system have Gaussian probability density functions at every point in time.

☒ **Correct**
Yes. This was assumed by the general Gaussian probabilistic inference solution derived earlier, and this assumption carries over to the Kalman-filter derivation.

2. What is the key result of the derivations performed in this lesson?

1 / 1 point

☐ They show how to determine values of process and sensor noise signals in real time from the measured input/output data from the system.

☐ They prove that the state estimate converges to the true system state as time progresses, even though the estimate may start with a large error.

☐ They prove that the process and sensor noise signals that act on the system are white and mutually uncorrelated.

☒ They show how to convert the abstract statistical expectation operations of generic Gaussian sequential probabilistic inference into matrix/vector equations that can be implemented in computer code.

☒ **Correct**
Yes. This is the main point of this lesson.

3. To compute $\hat{x}_k^-$, $\Sigma_{\hat{x},k}^-$, and $\hat{z}_k$, what information do we require? Select all that apply.

1 / 1 point

☒ We need to know model matrices A_{k-1} , B_{k-1} , C_k , and D_k .

☒ **Correct**
Yes. These matrices are needed in the prediction steps.

☒ We need to know the process-noise covariance matrix $\Sigma_{\bar{w}}$.

☒ **Correct**
Yes. We need to know this information for step 1b.

☐ We need to know the sensor-noise covariance matrix $\Sigma_{\bar{v}}$.

☒ We need to have an estimate of the prior values of the state vector, $\hat{x}_{k-1}^+$.

☒ **Correct**
Yes. This is needed in step 1a.

☒ We need to know the error covariance matrix of the prior state estimate, $\Sigma_{\hat{x},k-1}^+$.

☒ **Correct**
Yes. This is needed in step 1b.